
Pocket-sensitive somatic variant effect prediction (VEP) using AlphaFold-predicted protein structures

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[JR: abstract once results section is locked in.]

1 Background

Over the past twenty years, there have been a handful of recurrent cancer mutations that have rewritten how certain patients are clinically treated. BRAF V600E moved a significant chunk of metastatic melanomas off cytotoxic only regimens. EGFR exon 19 deletions and L858R allowed many with NSCLC to be managed with a daily tablet.

Cancer treatment now incorporates tumor sequencing and other biomarker assays to identify actionable alterations. Examples include BRAF V600E in melanoma, EGFR ex19 deletions in NSCLC, KRAS G12C across multiple tumor types, and HER2 amplification in breast/gastric cancers. While the promise of personalized medicine (matched therapy to mutational profile) has not fully reached its potential and conventional modalities of treatment (chemo, surgery, immunotherapy) remain bedrocks of cancer care, molecularly targeted therapies have transformed outcomes for particular subsets of patients with particular driver alterations. Variant interpretation, as a result, is at the crux of any clinical workflow, informing trial and therapeutic eligibility, treatment, and ultimately prognosis.

[JR: paragraph 2 – other approaches.]

[JR: paragraph 3 – our motivations / background into folding algorithms / importance in integration, uniqueness, hypothesis.]

2 Motivations & Hypotheses

[JR: write.]

3 Methods

3.1 Dataset

We downloaded the `variant_summary` from ClinVar which provides us with __ rows of Gene Variants and its Oncogenicity / Clinical Significance label [Landrum et al., 2018]. Using the Gene and Variant pairs, we can then produce a mapping to UniProt, yielding around 16k unique gene symbols, consisting of multiple isoforms.

For AlphaFold structure information, we first downloaded the AlphaFold human proteome, which contains __ pdb files [Jumper et al., 2021, Varadi et al., 2022]. And matching the gene and variant, we can extract the structure features from AlphaFold and pairing with external libraries such as fpocket [Le Guilloux et al., 2009] and DSSP [Kabsch and Sander, 1983].

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Figure 1: End-to-end pipeline. Variant labels are drawn from ClinVar and bridged to AlphaFold structures via UniProt; sequence, structure (fpocket + mkdssp + pLDDT), and evolutionary (phyloP / phastCons) feature blocks are joined into a single feature matrix, then evaluated under a gene-grouped split.

For Evolutionary information, ____ [JR: fill in.]

3.2 Feature Engineering

A full list of features that we included can be found in the Appendix.

Sequence Level. Amino Acid information: We extract Amino Acid mass, Kyte-Doolittle hydropathy index, and BLOcks SUBstitution Matrix (BLOSUM) from Biopython [Cock et al., 2009] and calculated the change in mass, hydropathy index, and charge as well to be incorporated as a sequence feature.

Structure Level. [JR: write.]

Evolutionary Level. [JR: write.]

3.3 Data Exploration

Figure 2 shows a subset of the features selected in all three feature sets, Sequence, Structure, and Evolution. For the sequence feature BLOSUM62, we see that the distribution is somewhat distinct in two classes where Oncogenic variants tend to have negative values and benign variants tend to have positive or 0 values, but there are still some overlap between the two. More obvious differences lies within Structure and Evolution features such as the pLDDT and SASA (Solvent-Accessible Surface Area), showing a more obvious difference in peak for Benign and Oncogenic. Similar behavior can be seen in phyloP100way feature from the Evolution set. These indicate that Structure and Evolution set shows distinct features and correlation with the oncogenicity/pathogenicity of the variant, and it can potentially be good predictors.

3.4 Model Choice & Evaluation

We chose to evaluate our data on 3 different models to ensure a wide variety of model types are covered, and the effects of feature set selection are fully studied. We chose to implement Random Forest, MultiLayer Perceptron (Neural Network, MLP), and XGBoost. For Random Forest, we use sklearn’s RandomForestClassifier with 300 estimators with no max depth restriction and a minimum sample leaf of 2, we also include the class weight as balanced [Pedregosa et al., 2011]. For MLP, we use sklearn’s MLPClassifier with two hidden layers of size 32 and 16 respectively and a max iteration of 500 [Pedregosa et al., 2011]. And for XGBoost, we use XGBClassifier from the

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Figure 2: Distributions of a representative subset of features across the three feature blocks (sequence, structure, evolution), split by class label.

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Figure 3: Headline macro-F1 across feature configurations and models. Bars show the mean over 5 seeds; error bars are ± 1 std.

xgboost library with 100 estimators, max depth of 4, learning rate of 0.1, `scale_pos_weight` of 1, and log loss as the evaluation metric [Chen and Guestrin, 2016].

3.5 Class Distribution

The class label distribution is mildly imbalanced ($\approx 1.4:1$ benign:oncogenic), so class balancing techniques are not strictly necessary. Nonetheless, we applied class reweighting to Random Forest and XGBoost models, as it is natively supported through `class_weight` and `scale_pos_weight` inputs, while keeping MLP models trained on natural distribution, as MLPs do not natively support class imbalance. The impact of class imbalance is expected to be small.

4 Results

A full table of the results for the 7 feature set combinations and 3 models can be found in the Appendix (Table 1).

[RW] figures/feature_lift.png
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Figure 4: Lift over the best constituent single modality, for each multi-modal feature set and each model. Positive lift indicates the modalities are complementary rather than redundant.

4.1 Macro F1 Analysis

- All combined has best performance, achieving highest F1 Macro score on all 3 models with stable std, not too much deviation across 5 seeds.
- 2 feature sets all outperform single feature sets, this indicates that it is not just one dominating feature set, but incorporating more features do improve classification performance.
- Training also becomes more stable as we see smaller standard deviation ranges as features increase.

[JR: tighten the bullets above into 1-2 paragraphs of prose for the final draft.]

4.2 Lift Graph Analysis

Lift means comparing with best single feature set performance, the improvement gained. So Sequence + Structure means $(\text{Sequence} + \text{Structure}) - \max(\text{Sequence}, \text{Structure})$.

- All combine yields highest lift in Random Forest, Random forest seems to benefit the most from additional features.
- Evolution only yields less improvement, it might be expected since there's only 2 new features.

“Across all multi-modal combinations and all three models, the lift over the best constituent single modality was positive (+0.03 to +0.12 macro-F1), indicating sequence, structure, and evolutionary features carry complementary signal.”

4.3 Feature Lift

For each multi-modal feature set, we compute

$$\text{lift} = \text{macro_f1}(\text{combined}) - \max(\text{macro_f1 of its constituent single modalities}).$$

For each of the 4 multi-modal sets (seq_struct, seq_evo, struct_evo, all), per model:

- seq_struct / RF: $0.814 - \max(\text{sequence} = 0.684, \text{structure} = 0.723) = 0.814 - 0.723 = +0.091$.
- all / RF: $0.863 - \max(0.684, 0.723, 0.739) = +0.124$.

[JR] figures/case_study.png
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Figure 5: Case study: a variant where the structure block flipped the prediction. The residue (red) sits at the edge of an fpocket-predicted binding pocket (blue mesh) on the AlphaFold structure.

What positive vs zero lift means.

- Lift > 0 : combining genuinely adds information beyond the best single ingredient – the modalities are complementary.
- Lift ≈ 0 : combining gives you nothing you couldn't get from just using the best single feature alone (modalities are redundant).
- Lift < 0 (didn't happen in our data): combining hurts – the model is confused by the extra dimensions.

A single grouped-bar figure showing all 12 lifts (4 multi-modal sets $\times$ 3 models) all positive is direct evidence of complementarity. Without it, a reader could counter “maybe all is just evolution doing all the work” – feature lift forecloses that interpretation by subtracting out the best single component.

5 Discussion

Discussion Point 1: Feature interpretation. [JR: write.]

Discussion Point 2: Different model types affect performance less as more features are added.

“The model gap shrinks as feature richness grows. The headline finding is feature-driven, not model-driven. Across all three model classes, adding structure and evolutionary features yielded a consistent $+0.17$ – 0.20 macro-F1 improvement, while differences between models within a feature set were ≤ 0.015 on the richer feature sets – within seed-to-seed noise. Model choice mattered most on the impoverished sequence-only setting, where MLP lagged tree-based models, suggesting tree models are more robust under low-signal conditions.”

6 Conclusion

[JR: short paragraph.]

References

Melissa J. Landrum, Jennifer M. Lee, Mark Benson, Garth R. Brown, Chen Chao, Shanmuga Chitipiralla, Baoshan Gu, Jennifer Hart, Douglas Hoffman, Wonhee Jang, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.
- Mihaly Varadi, Stephen Anyango, Mandar Deshpande, Sreenath Nair, Cindy Natassia, Galabina Yordanova, David Yuan, Oana Stroe, Gemma Wood, Agata Laydon, et al. AlphaFold protein structure database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research*, 50(D1):D439–D444, 2022. doi: 10.1093/nar/gkab1061.
- Vincent Le Guilloux, Peter Schmidtke, and Pierre Tuffery. Fpocket: an open source platform for ligand pocket detection. *BMC Bioinformatics*, 10(168), 2009. doi: 10.1186/1471-2105-10-168.
- Wolfgang Kabsch and Christian Sander. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22(12):2577–2637, 1983. doi: 10.1002/bip.360221211.
- Peter J. A. Cock, Tiago Antao, Jeffrey T. Chang, Brad A. Chapman, Cymon J. Cox, Andrew Dalke, Iddo Friedberg, Thomas Hamelryck, Frank Kauff, Bartek Wilczynski, and Michiel J. L. de Hoon. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422–1423, 2009. doi: 10.1093/bioinformatics/btp163.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

A Full results

B Full feature list

[JR: full feature listing per block.]

Table 1: Test split, mean $\pm$ std across 5 seeds. Bold = best feature set within each model (per metric); lower = better for ECE.

Feature set	Model	Macro F1	PR-AUC	ROC-AUC	Bal. Acc.	ECE
Sequence	Random Forest	0.684 $\pm$ 0.018	0.654 $\pm$ 0.041	0.747 $\pm$ 0.018	0.688 $\pm$ 0.020	0.074 $\pm$ 0.013
Sequence	MLP	0.651 $\pm$ 0.027	0.651 $\pm$ 0.039	0.746 $\pm$ 0.018	0.650 $\pm$ 0.025	0.038 $\pm$ 0.013
Sequence	XGBoost	0.682 $\pm$ 0.017	0.656 $\pm$ 0.041	0.748 $\pm$ 0.018	0.685 $\pm$ 0.017	0.076 $\pm$ 0.013
Structure	Random Forest	0.723 $\pm$ 0.034	0.718 $\pm$ 0.027	0.806 $\pm$ 0.037	0.722 $\pm$ 0.035	0.042 $\pm$ 0.028
Structure	MLP	0.736 $\pm$ 0.039	0.739 $\pm$ 0.035	0.817 $\pm$ 0.044	0.737 $\pm$ 0.040	0.059 $\pm$ 0.019
Structure	XGBoost	0.739 $\pm$ 0.037	0.738 $\pm$ 0.033	0.822 $\pm$ 0.042	0.744 $\pm$ 0.041	0.070 $\pm$ 0.019
Evolution	Random Forest	0.739 $\pm$ 0.020	0.702 $\pm$ 0.027	0.804 $\pm$ 0.022	0.741 $\pm$ 0.021	0.104 $\pm$ 0.014
Evolution	MLP	0.775 $\pm$ 0.026	0.762 $\pm$ 0.029	0.848 $\pm$ 0.017	0.774 $\pm$ 0.027	0.037 $\pm$ 0.006
Evolution	XGBoost	0.776 $\pm$ 0.016	0.776 $\pm$ 0.036	0.855 $\pm$ 0.018	0.781 $\pm$ 0.016	0.065 $\pm$ 0.006
Seq + Struct	Random Forest	0.814 $\pm$ 0.008	0.853 $\pm$ 0.011	0.896 $\pm$ 0.009	0.816 $\pm$ 0.008	0.028 $\pm$ 0.006
Seq + Struct	MLP	0.801 $\pm$ 0.005	0.843 $\pm$ 0.009	0.890 $\pm$ 0.008	0.801 $\pm$ 0.011	0.040 $\pm$ 0.028
Seq + Struct	XGBoost	0.813 $\pm$ 0.010	0.856 $\pm$ 0.009	0.897 $\pm$ 0.008	0.819 $\pm$ 0.009	0.047 $\pm$ 0.011
Seq + Evo	Random Forest	0.795 $\pm$ 0.012	0.818 $\pm$ 0.028	0.880 $\pm$ 0.013	0.798 $\pm$ 0.012	0.058 $\pm$ 0.009
Seq + Evo	MLP	0.807 $\pm$ 0.016	0.832 $\pm$ 0.021	0.893 $\pm$ 0.014	0.811 $\pm$ 0.016	0.024 $\pm$ 0.008
Seq + Evo	XGBoost	0.813 $\pm$ 0.014	0.846 $\pm$ 0.025	0.898 $\pm$ 0.013	0.820 $\pm$ 0.014	0.053 $\pm$ 0.006
Struct + Evo	Random Forest	0.819 $\pm$ 0.018	0.839 $\pm$ 0.022	0.899 $\pm$ 0.016	0.820 $\pm$ 0.017	0.022 $\pm$ 0.005
Struct + Evo	MLP	0.803 $\pm$ 0.022	0.826 $\pm$ 0.030	0.888 $\pm$ 0.022	0.808 $\pm$ 0.024	0.047 $\pm$ 0.023
Struct + Evo	XGBoost	0.818 $\pm$ 0.019	0.848 $\pm$ 0.020	0.902 $\pm$ 0.016	0.825 $\pm$ 0.019	0.046 $\pm$ 0.017
All Combined	Random Forest	0.863 $\pm$ 0.014	0.901 $\pm$ 0.016	0.936 $\pm$ 0.011	0.866 $\pm$ 0.012	0.030 $\pm$ 0.005
All Combined	MLP	0.854 $\pm$ 0.014	0.892 $\pm$ 0.018	0.930 $\pm$ 0.012	0.855 $\pm$ 0.014	0.021 $\pm$ 0.007
All Combined	XGBoost	0.857 $\pm$ 0.014	0.904 $\pm$ 0.011	0.936 $\pm$ 0.009	0.864 $\pm$ 0.013	0.040 $\pm$ 0.012